Unravelling Complexity: Linking Deep Neural Networks to Phenomena

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Abstract

The profound influence of Artificial Intelligence on society is more evident than ever before, extending its reach into various domains, including scientific inquiry. Machine Learning models become more accurate predictors, but at the same their workings become increasingly challenging to comprehend. This gives rise to a critical question; can we use models to understand phenomena if we don't understand the models themselves? Emily Sullivan argues that we can, making a case for 'link uncertainty'. This paper analyses this argument from both a technical- and philosophical perspective. Suggestions are presented to improve the concept of link uncertainty. The results aim to facilitate fruitful further discussions in the realm of AI in scientific practise.

Introduction

Artificial Intelligence (AI) is currently the centre of attention in multiple domains such as healthcare, finance and education. With the rise of publicly available AI-tools, like ChatGPT, we seem to be at the start of a new era. The velocity at which major developments have been made is a double-edged sword: a new world of possibilities opens up but at the same time we are unaware of the damaging consequences the technology might bring. Elizer Yudkowski (2008) has warned more than a decade ago for the catastrophic consequences AI models could have on society. Especially as we progress to a society which is more and more built upon AI, the call for transparency and accountability of these AI systems becomes greater. One of the main difficulties in unpacking the workings of an AI system (and its possible adversities) is the complex nature of the mathematical models that underlie the system. Such models are said to be 'opaque' as it is hard to grasp how they work exactly. The intricacies that arise when unpacking a model's opacity is best captured by the term 'the Black Box Problem'. As a response to this problem the field of Explainable AI (XAI) has emerged, which aims to bridge the gap between human understanding and the AI algorithms.

The emergence of sophisticated AI models also impacts the ways in which scientific inquiry is done. An important question that arises is; can we use models to understand phenomena if we don't understand the models themselves? Emily Sullivan (2022) argues that indeed this is possible and proposes her idea of 'link uncertainty'. In Sullivan's paper she discusses a specific type of Machine Learning (ML) model, Deep Neural Networks (DNNs). Understanding phenomena through DNNs in light of Sullivan's work is the topic of this paper. I also discuss the context surrounding Deep Neural Networks, the Black Box Problem and scientific understanding. I hope to demonstrate that some points made by Sullivan require improvement and I make suggestions to do so. My critique highlights the way Sullivan uses some fundamental notions and how she believes a model can be linked to a phenomenon. My solutions therefore consists of an improved definition of opacity and a demonstration of how a model can be linked to phenomena in a way not sufficiently accounted for by Sullivan.

Overview

In Section I explain the general functioning of a Deep Neural Network. Although not extensive, this serves to get an overall idea of how these systems work and what makes them complex. Also, I discuss the way the Black Box Problem is conceptualised and I offer a framework to confront it. The first part of Section II gives context to Sullivan's argument for Link Uncertainty and provides its definition. In the second part of Section II I investigate how the argument is built up and offer critique to it. This is concerned with how Sullivan uses simple examples, which in my perspective do not account for DNNs. In section III the philosophical notions of understanding, explanation and opacity are discussed and applied to the work of Sullivan. The findings from section II and III lead me to make suggestions to improve the idea of link uncertainty in section IV. A conclusion based on the content of this paper presents itself in section V alongside with both a discussion of my work and hints towards possible future research.

I Foundational notions and concepts

Deep Neural Networks are computer models that drew inspiration from the way the human brain is organised. They are made up of several interconnected artificial neuronal layers arranged into three different kinds: an input layer, one or more hidden layers, and an output layer (see figure 1). Together, these layers process and change the input data to generate predictions or classifications.

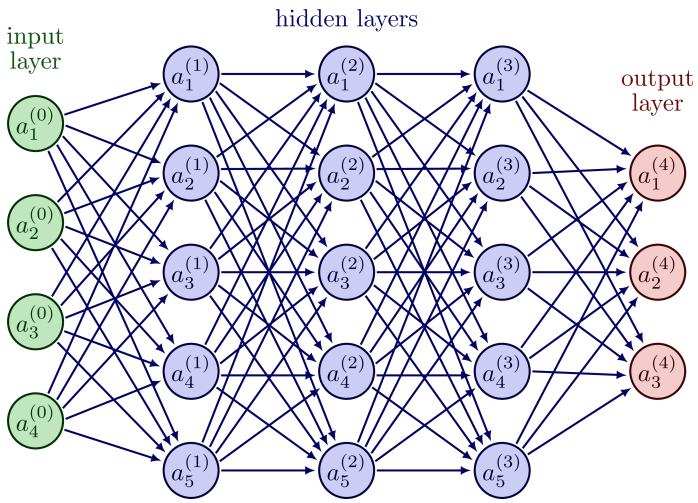

Figure 1: An example of a standard DNN architecture.

To add non-linearity and enable the network to recognize complex correlations in the data, DNN neurons apply activation functions to their inputs. The network receives input data during the forward propagation phase, and activations move through the layers as each neuron computes the weighted sum of its inputs, applies the activation function, and sends the result to the following layer. This nonlinear behaviour is in part what makes getting a hold of what the model does precisely a complex task. A technique known as backpropagation is employed to 'train' a DNN. Training a DNN means to make sure the parameters that the model settles on are optimal for the task it is assigned to. These parameters are the 'weights' connecting all the neurons from one layer to the neurons in the subsequent layer. These weights correspond with the lines between the different neurons (see figure 1). One can interpret these weights as an indication of to which extent the model needs to preserve certain information of the input data, to produce the best possible prediction. A loss function is used in this procedure to measure the difference between the predicted and desired outputs. The backpropagation algorithm finds the optimal weights through an iterative process designed to minimise the loss function. Since there can be many different neurons, connected through numerous weights, this process is high-dimensional which adds to the complexity of the model. During this process, multiple optimization approaches are used in addition to backpropagation to improve the training process. A DNN's performance can be tested using test data once it has been trained. When making predictions, fresh data is fed into the trained network, and the output layer shows predictions or categorises the data based on the discovered associations and patterns. Different kinds of DNNs exist to fit different needs, image classification is for example best realized by a different DNN architecture than the one needed for speech recognition. I take on a general notion of DNNs as their overall conception and the problems, with respect to opacity and understanding, that arise from them are similar (Sullivan, 2022, p.118).

Notably, these types of models are complex as they operate in high-dimensions and take on non-linear relations. Therefore, it can not be easily inferred what the model has done to come to the produced output. This is what we call the Black Box Problem in Artificial Intelligence and specifically this plays a role within Machine Learning. Machine Learning is a sub-domain within AI

which deals with models that 'learn' themselves to find the right parameters, as we just saw with DNNs. A central notion is that of 'opacity' which deals with the complex nature of these Machine Learning models. What makes models opaque is the difficulty to 'look inside' the model and explain the ways it has transformed input data into output. We see that increased opacity causes more difficulties in explaining what a model does and how it does so. Opacity has several consequences for the future use of these models (Zednik, 2021, p.266), an example can be found within the legal realm surrounding this technology. As pointed out by Goodman and Flaxman (2016), under the European Union's new General Data Protection Regulation, people that are subject to the decision of an automated system have the 'right to an explanation'. Although this right is currently hard to define precisely, it does suggest that for models to act as decision-makers they need to match certain criteria with respect to explainability. This means that when a model produces a certain output and thereby comes to a decision, someone affected by this decision could demand an explanation of how the model came to its output and decision. What one therefore needs to understand is how the input data, such as personal income and level of education for example, is transformed into the generated output, which could be the decision whether someone is eligible for a loan. A similar call has been made by Yoon et al. (2022) for the medical sector, where Machine Learning models are becoming increasingly important in high-stakes situations[1].

Multiple frameworks have been offered to get a hold of this problem (Beisbart & Räz, 2022, p.6), under which one from Zednik (2021). Zednik uses a specific definition of opacity, which will be discussed later, and adopts a stakeholder taxonomy offered by Tomsett et al. (2018). From this taxonomy we learn that within a certain ecosystem surrounding a Machine Learning system, we may find different stakeholders who interact differently with the model at hand. While a creator is mostly interested in understanding the technicalities to make the model work properly, someone affected by the model might have moral concerns to how his/her data is collected and used. Zednik (2021) expands on this idea by applying a levels of analysis framework originally developed by David Marr[2] to the Black Box Problem. Zednik makes the same distinction between three levels: computational, algorithmic and implementation. What this means is that when one asks a question regarding a Machine Learning model we can categorize the question in one of the three levels, where each level corresponds to a certain level of abstraction of the model. For example, the computational level is concerned with the behavior of the model in terms of how input is generally transformed into output while the implementation level deals with questions regarding the physical location of the computation, like the type of computer hardware. Combining this levels of analysis framework with the taxonomy of different agents surrounding a Machine Learning ecosystem, we arrive at Zednik's framework in which different stakeholders hold different kinds of questions regarding a model. This makes the Black Box Problem is not a monolithic concept but makes it rather dependent on the agent who is confronted with the model. Thinking of the Black Box Problem in this way makes it more definable and I aim to use this framework in the further sections.

[1] In their paper, Yoon et al. (2022) address 'interpretability' instead of explainability. The notion of interpretability is close to that of explainability although their exact definitions are hard to give (Beisbart & Räz, 2022, p.6)

[2] This framework was originally applied to the field of cognitive science and computational neuroscience, with the aim of understanding how information is processed and represented in the human brain.

II Link Uncertainty

Context and Definition

Sullivan describes that scientists can make use of models and algorithms to explain phenomena in the outside world (Sullivan, 2022, p.110). Her paper takes the perspective of models being 'part of explanations that are answers to questions that enable understanding of the target phenomenon' (Sullivan, 2022, p.111). Her description of explanation is that it aims at understanding (Sullivan, 2022, p.110). The context surrounding this description is discussed in section 3 together with the notion of understanding. Sullivan is specifically interested in using a model to understand a phenomenon as opposed to understanding the workings of the model. A similar distinction is made by Zednik (2021), in which understanding the model itself would correspond to the algorithmic/implementation level. Asking questions of *why* the model does certain things would correspond to the computational level. Utilising the framework proposed by Zednik, we can argue that a scientist is a specific agent which has certain needs of the model. In this case, the scientist is interested in using the model to give an explanation for a phenomenon he/she does research on.

Sullivan distinguishes between 'how-possibly' and 'how-actually' explanations we might yield in answering these *why* questions with regards to a phenomenon (Sullivan, 2022, p.112). How-possibly explanations highlight possible causal relations that might aid in explaining a phenomenon while how-actually explanations focuses on providing an actual explanation of the observed phenomenon. The ML model, like a DNN, we use can aid us in coming up with these types of explanations. As described previously, it is commonly thought that the opacity of the model makes understanding and explanation difficult. Sullivan on the other hand believes this is not quite the case and argues that 'model simplicity and transparency are not needed for understanding phenomena' (Sullivan, 2022, p.110). Instead, it is the level of 'link uncertainty' that limits possible understanding she argues.

Link uncertainty is 'the lack of scientific and empirical evidence supporting the link that connects the model to the target phenomenon' (Sullivan, 2022, p.112). With the example of Schelling's model for segregation, Sullivan illustrates this concept. This is a computational model that demonstrates how simple individual preferences for similar neighbours can lead to segregation patterns in a population. Sullivan concludes from this example that the specific workings of the algorithm are not necessary for enabling understanding (Sullivan, 2022, p.116). It is rather the extent to which the algorithm simulates the real world that is important for explanation, which in turn would lead to understanding of the phenomenon. Whether the algorithm does a good job in simulating the real world phenomenon is determined by the empirical evidence that links the model to the target phenomenon, which Sullivan calls the link uncertainty. The removal of link uncertainty, thereby establishing a more direct link from the model to the phenomenon, would make understanding possible. The way she builds up her argument however is not resistant to critique and I discuss this in the next section.

Analysis of the argument

To make the case that it is link uncertainty that hinders our understanding, Sullivan introduces the concept of 'implementation black boxes'. This idea is used by Sullivan in a simple case, to thereafter use it with regards to DNNs. First, I explain her reasoning in the simple case and afterwards I demonstrate how this does not hold in the case of DNNs.

Implementation black boxes arise when 'some level of detail regarding how the model is implemented is obscured' (Sullivan, 2022, p.114). Sullivan describes this to be one of the ways in which a black box problem can occur as an implementation is unknown to the person dealing with the model. Sullivan gives the example of the computation of a factorial [3] to demonstrate what

[3] The factorial of a non-negative integer n is the product of all positive integers less than or equal to n. This is denoted as $n!$. Example: $5! = 5 * 4 * 3 * 2 * 1 = 120$

she means with this concept. Moreover, Sullivan tries to show that this type of implementation black box does not hinder any possible understanding. By comparing two different computational methods that yield to the same outcome, recursive and iterative, Sullivan shows that indeed for the calculation of factorials the chosen mathematical method does not make an important difference [4]. Therefore Sullivan concludes that understanding the exact implementation is not important, as the output is still the same and that is what we use to understand the phenomenon. Sullivan then proceeds to apply this idea to Schelling's model and she concludes that 'implementation back-boxing in itself does not undermine our ability to explain or understand phenomena.' (Sullivan, 2022, p.115). 'Back-boxing' is a term used by Sullivan to give a verb to the process of something that becomes a black-box. Because 'black-boxing' is normally used within the scientific domain surrounding AI, I use 'black-boxing' as opposed to 'back-boxing' in the remainder of this paper. Coming back to the concept of implementation black boxes, note that it is important as Sullivan uses the concept in the case of DNNs later on. Before she does, Sullivan explains that implementation black boxes can exist at different levels. These levels correspond to the depth at which details of a machine learning model are obscured. The computation of a factorial within a ML model would count as a low-level implementation, as this plays a small part in a bigger web of mathematical structures and computation. In the case of Schelling's model, Sullivan recognizes more mid-level implementation black boxes. This consists of an updating algorithm (Sullivan, 2022, p.116), which is a higher-level process than a single computation. Sullivan argues that these mid-level implementation black boxes do not hinder any understanding because the result of the algorithm is not dependent on a specific implementation. Therefore Sullivan argues that 'It is not implementation back-boxing that gets in the way of understanding, it is link uncertainty.' (Sullivan, 2022, p.116). Before moving on to DNNs, Sullivan considers the case of the highest possible level of implementation black boxes. In this scenario we would only have a mapping from input to output, while completely in the dark of every step in between. Sullivan argues that this would make understanding impossible as we are completely unaware of how our input data is transformed to output. Sullivan hints that when the model is not black-boxed completely, we deal with an implementation black box as described before (Sullivan, 2022, p.117). These types of implementation black boxes are not limiting to understanding, she argues. Up until this point Sullivan seems to be right about implementation black boxes and correctly applies this concept to the computation of factorials and to Schelling's model. However, when these ideas are applied to DNNs some problems make themselves apparent.

First of all, the concept of the 'highest level black box' does not seem to provide any solid ground to build an argument upon. Sullivan's reasoning goes as follows: (A) complete black boxes make understanding impossible, (B) DNNs are not complete black boxes so therefore (C) understanding from DNNs is possible. The theoretical situation of a complete black box, mapping directly from input to output, is unfortunately merely theoretical and also illogical. How can we deal with a model that transforms our input, without knowing a single thing about its workings? There is no model ever created from which all complexity is obscured, because the model that transforms the output has been constructed by someone, who would have to had access to the tools that build up the model. These tools would consist of mathematical functions, which in aggregate would form the architecture of the model and define the way it works. Therefore the idea that we are given a model from which every single detail is obscured is impossible. Even when the ancient Greeks went to the oracle of Delphi to have their questions answered, it seems like less black-boxing took place. Mythology aside, it seems unreasonable for Sullivan to write:

> So, while the modeller does not have direct control over the modelling process, a contrast case with Schelling's model, still the process is not back-boxed at the highest level, such that it would prevent understanding of the phenomenon the resulting model aims to capture. (Sullivan, 2022, p.122)

[4] Implementations are of great importance when constructing a Machine Learning model, also when the same output is generated. This has to do with computing time for example, which plays a crucial role for a model's valuation

For the simple reason that complete black-boxing is not present in any case, because someone had to build the model which would require knowledge of some of the steps in between input and output, this feature of the argument seems redundant. Sullivan is correct that DNNs are not completely black-boxed, but because no model is it seems invalid to give this argument the weight Sullivan does. Setting this initial worry aside, it is more interesting to see how Sullivan further builds up her argument. The DNN case that forms Sullivan's example is that of the Deep Patient model from Miotto et al. (2016) This model has a DNN structure and 'takes as inputs large amounts of patient medical data and gives as an output a generalizable patient representation that can be used to predict future medical problems' (Sullivan, 2022, p.109).

Sullivan writes about the makers of the Machine Learning models, while explaining the technical details of DNNs, that they are 'not working completely in the dark' (Sullivan, 2022, p.121). The fact that these modellers rely on common statistical knowledge is enough for Sullivan to state that understanding is not prevented by having implementation black boxes as described before. I argue that this reasoning does not hold, especially not when investigating the assumption that underlies the argued unimportance of implementation black boxes for limiting understanding. Indeed, when we consider two methods that yield the same output (like in the example of the factorials) the specific implementation does not make a difference. But present in this example is the assumption that the different implementations yield **the same** output. This assumption is also taken when Sullivan discusses Schelling's model, stating that the implementation does not matter to the general output pattern (Sullivan, 2022, p.116). While the assumption might be true for Schelling's model, this is not that straightforward for DNNs. If we consider a low-level implementation of a DNN, let's say the activation function, we do see that it is important to grasp its complexity. Choosing a specific activation function causes the model to behave differently and therefore change the possibility of understanding. An example presents itself when looking at a specific activation function, the ReLu[5] function. As Nwankpa et al. (2020) explain, multiple slightly different versions of this activation function exist. One of the reasons this is the case is that sometimes during the training phase of a DNN, some neurons 'die' (Nwankpa et al., 2020, p.9). When this happens this neuron will not be activated and this alters the model's capability of making accurate predictions and the overall way in which information is represented. While this does not necessarily mean that the overall general output becomes radically different, the mere possibility that this could happen is enough to state that understanding the specific implementation of the function is important. For my example, if a modeller uses a ReLu function he needs to be cautious about which specific implementation of this function he uses. Sullivan herself writes:

> If the low-level implementation details made a difference to the high level results of Schelling's model, then the implementation would matter for explaining and understanding segregation. (Sullivan, 2022, p.115)

Therefore, it seems that Sullivan's simple examples are not adequate to account for DNNs as within these models low-level implementation details do make a difference to the output. Beisbart & Räz also note the argument made by Sullivan:

> One crucial argument for her claim is as follows: Black boxes need not compromise the scientists' abilities to use a model if they merely black-box the implementation of the fulfillment of a known task (e.g. the calculation of a factorial). (Beisbart & Räz, 2022 , p.5)

In Beisbart and Räz's interpretation of Sullivan, black-boxing the fulfilment of a known task does not lead to problems for scientists to use the model. However, one can spot how this does not apply to DNNs. As for DNNs we do not only have simple implementation black boxes for tasks we know how to fulfil, but also implmentation black-boxes from which we do not know how the task

[5]ReLu stands for Rectified Linear Unit and is used in numerous DNN architectures because of its simplicity and reliability (Nwankpa et al., 2022, p.9)

is fulfilled. This is for example the mid-level implementation of updating the weights in the DNN after each iteration during the training phase (Sullivan, 2022, p.122). The updating of the weights is a high-dimensional process which we don't exactly understand in terms of how it is done. In Beisbart and Räz's formulation, here we find a task that is black-boxed from which the fulfilment we do not know. Therefore it seems that also in their interpretation, Sullivan's argument does not apply to the DNN case.

A possible counterclaim that Sullivan could make is that the choice for a specific low-level implementation, like the activation function, is accounted for in her explanation. Because she mentions that modellers rely on common statistical knowledge to make 'generalizable' models (Sullivan, 2022, p.121). The thing to be commented is that this claim does not dispute my previous argument, because two models which aim to predict the same phenomenon can both be generalizable although they do not produce the same output. Secondly, if this kind of statistical knowledge is not included in understanding the model's opacity, then what is? This is an important question as the argument depends on how you define the opacity of the model and how this would not limit understanding. This I handle in the next chapter when taking a closer look at the epistemological elements of Sullivan's argument.

III Philosophy of XAI

Centrally present in the current discussion on link uncertainty are notions of understanding, explanation and opacity of a DNN. Philosophy of Science has traditionally dealt with the former two and this therefore forms a sensible point of departure when discussing the epistemological elements of the Black Box Problem. In this section I present the way Sullivan deals with these concepts and how that relates to the argument she makes for link uncertainty. Sullivan writes about the main aim of her paper:

> The central issue in this article is whether explanations that utilize complex and opaque models are unable to provide understanding of phenomena in virtue of the fact that the model itself is not well understood or back-boxed. (Sullivan, 2022, p.111)

Therefore it seems reasonable to have a closer look to which extent the model is not well understood according to her. Moreover, it might be interesting to see which level and what kind of understanding is tacitly included in constructing models in the first place. I start by looking at Sullivan's ideas with regards to understanding and afterwards discuss the problem of opacity.

In fact, Sullivan explicitly evades any discussion on what is meant by understanding (Sullivan, 2022, p.111). She does write that she takes on the perspective that 'explaining why helps us to understand why' (Sullivan, 2022, p.110). Using the terminology provided by Baumberger et al. (2017) this seems to be of the 'explanatory' understanding type. Explanatory understanding is different from objectual understanding in that it characterises itself to deal with *why* something is the case. Objectual understanding leads someone to understand a subject matter or domain of things, which does not necessarily involve any why-questions. It is good to mention again that Sullivan describes models to be vehicles to our understanding of phenomena. As Baumberger et al. (2017) describe, an account of understanding ideally would state the necessary and sufficient conditions for understanding to take place within someone. The most interesting one for the present discussion is the condition of factivity. It is not of use to test this condition for the understanding Sullivan aims to obtain (of the phenomena) but rather the understanding she assumes to be present in the modellers. Sullivan does not explicitly state which understanding modelers possess but she does write that 'the modeller relies on a wealth of knowledge and research about what methods to follow to build a generalizable model for the task at hand' (Sullivan, 2022, p.122).

Besides that, Sullivan writes that

> different types of saliency testing are enough to satisfy our need to know the high level details of how the model works to open the door to understanding the phenomenon the model bears on (Sullivan, 2022, p.122)

These types of saliency testing refer to post-hoc analytic tools to uncover which input features are most responsible for the output (Zednik & Boelsen, 2022, p.223). These XAI tools are important in the world of Explainable AI, as they seek to increase our understanding of how a model came to its derived output. Therefore, XAI tools are seen to help in resolving the Black Box Problem[6]. An example (See Figure 2) in the realm of image classification is that of input heat mapping, which shows which input regions of a photograph carried the most responsibility for the model's output. While Sullivan mentions that these techniques might enable the possibility of understanding a phenomenon, I argue that these techniques (and our understanding of them) are in fact precisely what is needed to obtain understanding of a phenomenon. I discuss this in section 4 but first I focus on the factivity debate around these techniques, as they are also important in Sullivan's argument.

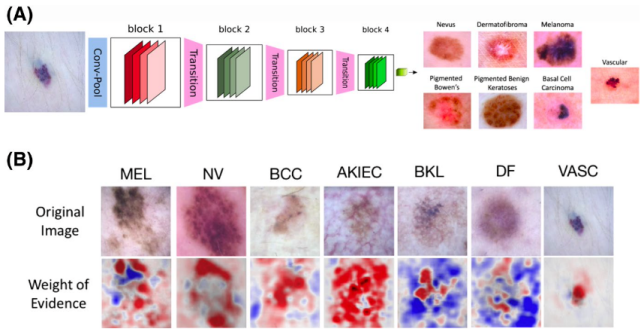

Figure 2: Taken from Zednik & Boelsen's (2022) paper: 'Reproduced from Li et al. (2019). A (top): A convolutional neural network, trained to detect and classify cancerous melanoma from images of skin discoloration. B (bottom): Input heatmaps depicting high-responsibility (red) and low-responsibility (blue) input regions for specific classifications'

The reason we need these post-hoc methods, which are idealisations of the true model, is that many DNNs are too complicated to be completely understood (Beisbart & Räz, 2022, p.7). But as soon as one uses an idealisation to enable understanding, philosophical awareness is called for. Although controversial to spell out, there exists a consensus that a representation must be compatible with facts to a certain extent (Baumberger et al., 2017, p.6). In this discussion we find the strong claim for factivity, meaning that all propositions constituting an explanation must be true. This claim is in general too extreme; we see in practice that falsehoods are present in almost all explanations while we do state to possess understanding of the object in question. An example is Corpernicus' astronomical model, which definitely enabled greater understanding on the motion of planets but still included the falsehood of planets moving in circles rather than ellipses. Therefore we find a weaker camp with regard to factivity, the so-called moderate factivists (Baumberger et al., 2017, p.6). Their claim goes that not all, but only the central propositions leading to the explanation need to be true. What it means for a proposition to be 'central' is not part of this paper. It is however interesting to see how these ideas apply to idealisations of a DNN, which are used to constitute an explanation. The opponents of moderate factivity, so-called non-factivists, believe that even moderate factivity is too bold of a condition. Their claim would be that if idealisations

[6]Note there is also another camp within XAI that deals not with post-hoc explanations to solve the Black Box Problem, but rather tries to find ways to evade problems regarding understanding at all (Zednik, 2021, p.266).

are the best science can currently do, the lack of factivity is in fact compatible with understanding, to a certain extent (Baumberger et al., 2017, p.7). In our case that would mean that post-hoc methods, given that they are our best option at the moment, are in fact able to provide understanding. Baumberger et al. write about the proponents of moderate factivity they would argue that

> Understanding a target system in terms of an idealized model requires that one knows what the model is like and how it relates to the target. This, in turn, involves knowing to what extent the model is an idealization, which aspects of the target it is intended to capture, and under which conditions the target approximately behaves as the model. But if a scientist knows all this, her central beliefs about the target are true. (Baumberger et al., 2017, p.8)

This argument is used to show that moderate factivity is preserved using idealised models. Interestingly we see that understanding of the model and which aspects of the target it is intended to capture are taken to be necessary to understand a target system. This adds to the claim I make in section 4 for XAI tools decreasing link uncertainty, hence enabling understanding. The exact philosophical position Sullivan takes in this debate is hard to infer, but there are some hints. It seems that Sullivan throughout the paper validates the use of DNNs for the virtue of them having high predictive accuracy (Sullivan, 2022, p.117). A DNN having high predictive accuracy says nothing about the factivity of the explanation the DNN provides. What we do find is that this idea is compatible with what De Regt and Gijsbers (2016) have proposed, namely an additional condition for understanding: scientific effectiveness. They write:

> scientific effectiveness is the tendency to produce useful scientific outcomes such as correct predictions, successful practical applications and fruitful ideas for further research (De Regt & Gijsbers, 2016, p.21)

De Regt and Gijsbers also write that to hold a non-factivists would require an argument which shows that nonsensical explanations do not lead to understanding. This means that if one does not think that explanation should be based on (a core of) facts, one has the challenge of showing that a false explanations does not lead to understanding. For example; someone argues that planets move around the sun in a square-like manner because he has seen this in his dream. Here we find a false theory but as we don't require theories to be based on facts we do need to show that this person's false and nonsensical explanation does not lead to understanding. Therefore, the condition of scientific effectiveness is proposed as a solution for holding a non-factivists standpoint (Baumberger et al., 2017, p.9). This means that a theory can still be justified as long as it has scientific effectiveness, e.g. making correct predictions. In the light of this information, I suggest that Sullivan takes a non-factivist standpoint while taking the predictive value of DNNs as a justification. As we now have a better idea of Sullivan's conception of understanding, it is necessary to see how she is concerned with what we supposedly do not know about a DNN. This concept is best captured by the term 'opacity', which is perhaps the main cause of the Black Box problem in general (Zednik, 2021, p.266). It is a good starting point to find out what kind of opacity Sullivan means. Here one needs to be cautious as Sullivan first writes that she is concerned with a term introduced by Paul Humphreys and refers to Humphreys' notion as:

> 'epistemic opacity': the extent to which the process of the model and its derived output are inaccessible to scientists and modellers. (Sullivan, 2022, p.111)

In the footnotes however, Sullivan writes that the opacity she intends to use is close to Burrell's use of opacity in regards to the way algorithms 'operate at the scale of application' (Sullivan, 2022, p.111). As Sullivan's conception of opacity is 'close' to the one Burrell describes, I present the quote which best captures Burrell's explanation of opacity:

> While datasets may be extremely large but possible to comprehend and code may be written with clarity, the interplay between the two in the mechanism of the algorithm is what yields the complexity (and thus opacity). (Burrell, 2016, p.5)

So here we see that the opacity arises in the interplay between written code and the analysed dataset. 'In the mechanism of the algorithm' seems to mean the moment when the model actually performs its computations, as then the code and the dataset engage with each other. This notion of opacity is compatible with the previous reference to Humphreys', in fact it makes it more specific. But the question remains whether Sullivan is actually using this type of opacity for making her argument. As we saw in section 2, Sullivan uses her concept of implementation black boxes, which would have to be compatible with her notion of opacity to make a good argument. We see that most of the 'simple' implementation black boxes are not compatible, because the main idea was that this meant that some details of the model were obscured. But as these details are found within the written code, not in the interplay between code and dataset, this type of opacity is not compatible with the one she intends to use. In the example of the computation of factorials, we see that it is the written code in which details are obscured and not in a certain interplay between code and data. Another part of her argument however discusses 'the ways in which the machine settled on certain weights for certain pieces of data given the complexity and high number of data points' (Sullivan, 2022, p.122). Here we do see a notion of opacity compatible with the one Burrell describes. This type of opacity does not cause any problems according to Sullivan, as it is not a complete black box (Sullivan, 2022, p.122). The concept of a complete black box is however not a valid one, as it is impossible to have a model which maps directly from input to output. Someone has had to make a model using mathematical methods, which would illuminate some of the steps in between input and output.

In conclusion, although Sullivan explicitly evades the discussion around understanding some inferences can be made. On a theoretical ground, she seems to hold a non-factivist position with taking the predictive value of DNNs as justification. Furthermore, Sullivan acknowledges that post-hoc techniques such as heat mapping are needed for the models to have explanatory power. Also, Sullivan's use of the notion of opacity is not consistent in all her examples. When she does apply her intended notion of opacity correctly, the explanation of why it does not hinder understanding is at least debatable. In the next section, I make suggestions on how to overcome the identified problems of her argument.

IV Suggestions for improvement

> Much more can be said on what precisely it takes for there to be an adequate link connecting a model to the phenomenon of interest. The cases discussed in this article indicate that the stronger the connection we have between the model and the phenomenon, the greater the understanding. (Sullivan, 2022, p.129)

In this section 'more will be said' about establishing a link between the model and the phenomenon at hand. The last sections dealt with the elements which make up Sullivan's argument and the inquiry into these elements. Now, I would like to make arguments to improve the concept of link uncertainty. To repeat, the core of Sullivan's paper is to show that it is not the model's opacity, but rather link uncertainty which limits understanding of a phenomenon. Link uncertainty is the lack of evidence that connects the model to the phenomenon. My argument is that to decrease link uncertainty using models, we do need to deal with the model's opacity. We could do so by uncovering the model's workings with the help of XAI tools, such as heatmapping. What is especially important in my perspective is that we should look 'within' the model to establish the link with a phenomenon. In this way we can obtain more understanding of how the model transforms input data into output. As the model aims to make correct predicts about a phenomenon, uncovering the way it came to its prediction can enable understanding of the phenomenon. This is in contrast to Sullivan, who argues for additional evidence that does not have to do with the model's workings per se. The further part of this section elaborates on my argument and its different aspects.

As we have seen in section 3, Sullivan writes that saliency testing procedures (the XAI tools) are needed to 'open the door to understanding'. While she seems to acknowledge the value of

these methods without paying further attention to it, other writers do focus more on the value XAI tools can bring. Zednik and Boelsen write about Sullivan's paper and agree that link uncertainty is indeed a threat to the use of Machine learning for scientific inquiry (Zednik & Boelsen, 2022, p.226). However, Zednik and Boelsen argue that XAI techniques are not merely a tool to enable the possibility of understanding, but can be used to combat link uncertainty and gain insight into the phenomenon at hand. Zednik and Boelsen write:

> Notably, although the ML model itself is responsible for identifying relevant skin samples, it is the heatmapping technique that allows scientists to determine, characterize, and interpret those features, and thus, to possibly refine their conception of the target phenomenon. (Zednik & Boelsen, 2022, p.227)

This quote shows that indeed the model itself is responsible for identification, which would accord with answering how-possibly questions in Sullivan's description. But, because the XAI techniques allow us to understand which aspects were of high-importance to the output, we can gain insight about the phenomenon it bears on. In Zednik and Boelsen's words: 'What emerges is a picture in which XAI techniques for identifying high-responsibility input features facilitate the task of refining target phenomena.'(Zednik & Boelsen, 2022, p.228). Zednik and Boelsen also write on the value XAI tools have in guiding future research, but this is not part of this paper. To see whether my argument of using XAI tools to decrease link uncertainty and thereby enabling understanding of phenomena is compatible with Sullivan, it is important to know what she proposes as evidence to constitute a link:

> The way of establishing the necessary link is with additional scientific evidence that supports the connection between the causes or dependencies that the model uncovers to those causes or dependencies operating in the target phenomenon. (Sullivan, 2022, p.113)

XAI tools can precisely do what Sullivan demands: connect the causes that the model uncovers to the causes present in the phenomenon. This is due to the fact that XAI tools give answers to why the model gives certain output, which is necessary to establish the link. To establish a link one needs to have an idea of how a model relates to the target phenomenon. If one therefore knows which parts of the input data are mostly responsible for making (accurate) predictions about the target phenomenon, one can uncover which parts of the model link to the phenomenon. Further support for my claim can be found in Burrell's text on opacity within Machine Learning. Burrell writes that one of the approaches to deal with opacity is ''feature extraction', an approach that analyses what features actually matter to the classification outcome, removing all other features from the model.' (Burrell, 2016, p.9). This is exactly what XAI tools do, namely making sure only the important features for the output get highlighted. When Sullivan gives a specific example of a possible type of evidence she writes:

> The type of scientific evidence needed to reduce the link uncertainty, in this case, could consist of building additional statistical models that makes the deep patient results more robust (Sullivan, 2022, p.125)

If we replace 'additional statistical models' with 'XAI tools' we see that my argument gains strength. We are sure we might do so as XAI tools are in fact additional models which not only make the original model more robust but also help to explain it. As the XAI tools create a stronger link between the model and the phenomenon, greater understanding is realised. The difference however is that XAI tools, serving as additional statistical models, are designed to look into the workings of the original model, opposed to statistical models outside of the original one.

Another point I want to raise is the use of the term 'opacity', as this deserves careful analysis (Zednik, 2021, p.268). While Sullivan refers to both the broad definition made by Humphreys and the more specific one from Burrell, I want to propose a definition which aims to solve the identified problems. The definition I refer to is also given by Humphreys and is spelled out as follows:

> epistemically opaque relative to a cognitive agent X at time t just in case X does not know at t all of the epistemically relevant elements of the process. (Humphreys, 2009, p.618)

Although this definition is quite broad, using Zednik's work on a normative framework for XAI gives us more grip. Zednik (2021) writes that from this definition flow two important takeaways; the first being that opacity is agent-specific and the second being that opacity concerns the lack of a certain knowledge. We would then need to spell out what an 'epistemically relevant element' (ERE) is for a specific agent. If we have done this correctly, explanation can be seen as the acquisition of these EREs (Zednik, 2021, p.269). As this type explanation serves a specific agent, this helps to combat the subjective nature surrounding the debate as noted by Beisbart & Räz (2022). In the case of using DNNs for scientific inquiry, this would mean that opacity consists of the unknown elements which would link the model to the phenomenon. The cognitive agent would then be the scientist and his aim with the model would be understanding the phenomenon.

Opacity in this definition is not an abstract concept but rather an invitation to postulate the specific EREs to a certain agent. These EREs then help in a more concise identification of the problem, which helps in solving it. To come up with different EREs we can use Zednik's framework to see which EREs are of importance to which agent. In Zednik's framework we have different agents who hold different epistemic needs with regards to a Machine Learning model. These different needs can be expressed in questions operating at three different levels, as we have seen in section 1. For example; the creator of a DNN model used for scientific inquiry might be interested in how- and where-questions, which are at the implementation level. These questions have to do with the algorithm itself and where the physical computation takes place (Zednik, 2021, p.278-279). Properties of the underlying hardware and its consequences for the model's computation would be sensible EREs for this agent. On the other hand, a scientist who wants to use a DNN for his research would most likely not have this concern but rather be interested in why-questions, which are at the computational level. These questions are of the sort that has been central throughout this paper, namely to uncover why the model gave a certain output. Overall, this framework approach to opacity leads to more concrete problem identification.

To check whether my suggestions have any direct effect on the theory of link uncertainty, I regard a specific concern Sullivan mentions at the end of her paper. Sullivan notes rightly that if it is additional scientific evidence that constitutes understanding, the model itself might not play any epistemic role at all (Sullivan, 2022, p.129). This feels like something to be avoided as our present models seem capable of incredible results. My view is that XAI tools, that are additional but do look into the original model, can in fact establish a link and thereby enlarge understanding. This would mean that the worry of Sullivan is solved, as we use the models to make predictions and afterwards use XAI tools that give explanatory power to the complex model.

A second worry Sullivan mentions is that of inductive risk, which refers to generalisations or predictions made from scientific inquiry that are potentially harmful. An example is a facial recognition system that is biased towards a certain ethnic group, which could lead to it making discriminatory decisions. The ethics surrounding Machine Learning models are indeed an important concern (Beisbart & Räz, 2022, p.5) and deserve great attention. I propose that the use of the term opacity in an agent-relative manner is a good step forward in addressing this concern. As we can define different EREs for different stakeholders, we are able to change the focus on what's important for different situations. For example, the user of a scientific model which recognizes cancerous cells is specifically interested in circumstances the model gives a false negative output. False negatives would be disastrous in medical situations where life and death are at stake. This type of situation would be in contrast to a scientific model that is trained to recognize self-identifying homosexuals from pictures [7]. Here a scientist's ERE would be not that specifically concerned when the model gives false negatives. In this case it might be more interesting for the scientist to uncover which elements of the picture play the largest role in producing the prediction.

[7] This study is from Wang and Kosinski (2017) and is also mentioned by Sullivan

With regards to the factivity debate, my suggestions seem to stand on the same ground as that of Sullivan. To require a non-idealized explanation of a model which takes in millions or billions of parameters seems implausible. Therefore, I argue that the loss of factivity should be taken for granted as it trades off with great predictive power. The call for the condition of scientific effectiveness laid out by De Regt en Gijsbers (2016) will probably become even greater, as Machine Learning models become even better predictors but also more complex.

V Conclusion & Discussion

This paper dealt with the argument made by Sullivan for link uncertainty from both a technical and a philosophical perspective. In doing so, I noted several flaws which require improvement. The first is that a complete black-box is not a valid concept, as there doesn't exist any model that has every single detail obscured. This is never the case as to build a model we would need to define the transformations we apply to our input, so a direct input-output correspondence does not exist. Secondly, the way Sullivan uses the concept of implementation black boxes to argue that those do not hinder understanding is not applicable to DNNs in the way she intends to do. Besides that, the greater deal of implementation black boxes she describes are not compatible with the type of opacity she intends to use. Besides analysis on the notion of opacity, I demonstrated what level of understanding Sullivan assumed to be present when constructing DNNs. Within this understanding we find that saliency testing, referring to XAI tools, are present but not any further discussed by Sullivan.

I presented my argument which would refute the claim that we do not need to address a model's complexity to understand phenomena. My argument goes that XAI tools, such as heatmapping, can be used to unravel the model's complexity. This unraveling would mean that a greater link between the model and the target phenomenon is established, thereby enabling understanding of the phenomenon. Multiple writers like Zednik, Boelsen and Burrell seem to be in agreement that XAI tools can decrease opacity, and Zednik and Boelsen (2022) specifically argue for XAI tools to enable understanding of the target phenomenon.

As the notion of opacity is of major importance in this discussion, I have suggested a definition from Zednik (2021) for future analysis. This definition is agent-relative and can be used to specifically define what an EREs is for different agents. By using this framework epistemic problems can be identified better and therefore a solution to them is closer. While Sullivan explicitly evades the philosophical debate surrounding understanding, I have tried to shed some light on this subject. I have done so by looking at the condition for factivity, which is specifically interesting for XAI tools. This is due to the fact that these tools are idealisations of the true model, and therefore factivity is partly lost. I proposed that another condition for understanding, that of scientific effectiveness, can be used to justify this loss.

These findings provide a possible improvement of concept of link uncertainty, but also a discussion of them is warranted. First of all, more work on philosophical understanding with regards to DNNs and Machine Learning in general is required. My suggestions are not comprehensive to the entire array of problems and should be considered as proposals to improve the work of Sullivan. This entails that there is definitely credit to the idea of link uncertainty, and also that Sullivan's proposed ways to decrease it are valid. The idea that additional empirical evidence can indeed establish a link between a model and a phenomenon seems more than reasonable. What can be further inferred from my analysis is that the relatively new field of AI is in need of further philosophical investigation. This gets clear from the fact that multiple writers use the same concepts with a different meaning. A possible action that can be taken for more consensus is to explicitly work out EREs in different domains. This could help agents like law-makers and regulators to create guidelines and boundaries that are necessary for safe use of this promising technology. Also, while I refer to XAI tools in general, more specific research can be done on the different variations and the values that they would offer in separate use-cases. In this paper, just like Sullivan, I have

also not dealt with the philosophical view on the relation between explanation and understanding. Working this out in light of the present discussion might prove itself to be fruitful. Besides that, I am curious to see how the concept of scientific understanding itself will change over time, as the idea of an hyper-intelligent entity does not seem implausible anymore. Many more interesting philosophical questions are sure to arise with the fast-paced development of AI. Therefore, a lot more intellectual work is needed and this paper can be seen as a (very) tiny step in that direction.

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